

Khaled Sid'El Vally

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PROFILE

Engineering student with hands-on experience building retrieval-augmented generation (RAG) systems, ML pipelines, and production-grade AI applications. Proficient in Python, vector databases, and deploying scalable intelligent systems from research to deployment. Seeking AI Engineer / Machine Learning Engineer opportunities.

EDUCATION

Higher School of Statistics and Information Analysis (ESSAI) <i>National Engineering Diploma in Statistics and Data Science</i>	Tunis, Tunisia Sep 2023 – 2026
IPGEI, École Supérieure Polytechnique <i>Preparatory Cycle for Engineering Studies (Math/Physics)</i>	Nouakchott, Mauritania 2021 – 2023

EXPERIENCE

Lynaat <i>AI Engineer Intern</i>	Jul 2025 – Aug 2025
- Engineered a Retrieval-Augmented Generation (RAG) chatbot for legal document search using LangChain and FAISS. - Designed and executed factorial experiments to optimize document segmentation (chunk size, overlap method), significantly reducing context loss. - Developed a statistical error classification model (Logistic Regression) to predict and categorize retrieval errors (False Positives vs. Negatives). - Deployed a performance monitoring dashboard using Streamlit to visualize retrieval metrics and optimization results.	

Technocolabs Softwares <i>Data Scientist Intern</i>	Remote Apr 2025 – May 2025
- Tuned supervised and unsupervised learning models using Scikit-learn, utilizing cross-validation to enhance prediction accuracy. - Performed feature engineering and data preprocessing on complex datasets to prepare pipelines for ML training.	

Smart MS SA <i>Data Analyst Intern</i>	Jun 2024 – Aug 2024
- Built interactive Power BI dashboards integrated with Odoo ERP to monitor operational KPIs and visualize business processes. - Conducted Exploratory Data Analysis (EDA) using SQL to identify data quality issues and support strategic decision-making.	

PROJECTS

AI-Powered Medical Chatbot (Full Stack RAG)	Sep 2025 – Oct 2025
- Developed a containerized medical assistant using FastAPI (backend), ChromaDB (vector store), and Google Gemini AI. - Implemented secure user management (JWT/Bcrypt) and persistent conversation memory to handle multi-turn medical dialogues. - Built a responsive React/TypeScript frontend with Server-Sent Events (SSE) for real-time AI streaming. - Orchestrated deployment using Docker Compose, ensuring environment consistency and scalability.	
Real-Time Vehicle Monitoring System	Sep 2024 – Dec 2024
- Built a computer vision system using YOLOv8 and EasyOCR to detect license plates and seatbelt violations with high accuracy. - Deployed the detection pipeline via a Flask web application for automated, real-time video analysis.	

TECHNICAL SKILLS

- Languages:** Python, SQL, Java, C, R
- AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, LangChain, FAISS, YOLOv8, NLP, Computer Vision
- Web Development:** FastAPI, Flask, Django, React (TypeScript), Streamlit, HTML/CSS
- Data & DevOps:** PostgreSQL, MongoDB, ChromaDB, Power BI, Docker, Git, Pandas, NumPy, Matplotlib

LANGUAGES

English: Professional Proficiency (B2) — **French:** Professional Proficiency (B2) — **Arabic:** Native